Warfarin: Management of Out-of-Range INRs

Objective:

To provide practical strategies supported by the best available evidence for managing out-of-range international normalized ratios (INRs) for patients on long-term warfarin therapy.

Background:

1. Warfarin can be challenging to manage due to its narrow therapeutic range, variable dose-response among patients and common interactions with drugs, diet, alcohol, and other factors.
2. In patients who are taking warfarin, thromboembolic events and bleeding are strongly related to the time in therapeutic range (TTR). This is the proportion of treatment time that the INR is within the target therapeutic range (typically 2.0 to 3.0 or 2.5-3.5 for patients with mechanical mitral valve replacements).
3. Clinicians may underdose warfarin due to a perceived greater risk of harm from bleeding associated with supratherapeutic INR values. However, subtherapeutic anticoagulation has been shown to increase the frequency and severity of thromboembolic events.
4. Good INR control, defined empirically as a TTR >60%, may be best achieved by appropriately addressing both high and low INR values using a consistent approach (paper-based dosing nomogram or computerized decision support program).

Approach to Out-of-Range INR Values:

1. For each out-of-range INR value, attempt to identify the cause. See the section below, “Summary of Common Causes of Out-of-Range INRs”.
   Questions to ask the patient include:
   - *Tell me what warfarin doses you’ve taken in the past 2 weeks.* The patient may have mistakenly taken a different dosage regimen than what was prescribed.
   - *Have you missed any doses in the past week? If yes, how many? How do you ensure that all doses have been taken? Do you use a calendar to record doses? A pill box?*
   - *Have you started or stopped any medication or supplements (prescription or non-prescription) recently? Any new antibiotics? Any dose changes of your medications? Do you take acetaminohen? How many per day?*
   - *How is your appetite? Have you been eating regularly? Have there been any recent changes in your diet?*
   - *Do you drink alcohol? If yes, how often and how much alcohol do you drink? Do you drink just on weekends?*
   - *Do you consumer cannabis products? This includes CBD. If yes, how mch and how often?*
   - *Overall, how has your health been? Any infection? Fever? Diarrhea? Flu? Recent cold?*

2. Determine if a change in dose needs to made at all, and if so whether a **one-time change in the dose** is all that is required or if a **change in the maintenance dose** is required or both.
   - A change in the maintenance dose should be considered if there are at least **two consecutive out-of-range INR values (in the same direction) in a patient with previously stable, in-range INRs and for which there is no identified temporary cause.**
   - A one-time change in the dose is appropriate for patients in which a transient cause is i

Managing Single Out-of-Range INR Values:

For patients with previously in-range INR values who present with a single slightly out-of-range
above or below the target range), there are two management options:

1. Continue current maintenance dose and repeat INR in 1-2 weeks, OR
2. Make a one-time dose change (increase or hold by ½ to 1 single dose) and resume current maintenance dose. Repeat INR in 1-2 week.

The specific approach is influenced by the magnitude of the out-of-range value, previous experience of similar values in the patient and whether the patient has strong risk factors for thrombosis/stroke or bleeding.

Maintenance Dosing Algorithms:

There are many warfarin dosing algorithms available. Healthcare professionals managing patients on warfarin should become familiar with one approach in order to develop experience and consistency in making dosing changes.

One example of an available computer decision support tool is INR Online® http://www.inronline.ca/ (may require a subscription)

The **Figure** below is a sample, paper-based Maintenance Dosing Algorithm published by the American Society of Hematology (2014) for the management of **non-bleeding** patients taking warfarin. This nomogram serves as a guide and does not replace clinical judgment.

Figure: Warfarin Dose Adjustment in Non-bleeding Patients by % Dose Change of Total Weekly Dose (adapted from Cushman et al, 2014)

This algorithm is based on making a dose change as a percentage of the total weekly dose.

For patients with INRs of >4.5 but <10 and without clinically relevant bleeding, temporary cessation of VKA alone without the addition of vitamin K is suggested. Vitamin K may be given if INR >10, even in the absence of bleeding, depending on individual patient circumstances (e.g. risk factors for bleeding, risk for thrombosis if over-correction of INR, ability to have repeat INR testing).

Summary of Common Causes for Out-of-range INRs:

Table 1 and **Table 2** summarize common causes and management strategies for LOW and HIGH INRs, respectively. Ensure patients' ongoing understanding and potential causes of out-of-range INRs. When adding or discontinuing medications for a patient receiving warfarin, consider use of an evidence-based drug interaction program or reference, if available.

Table 1: Common Causes of LOW INRs and Management Strategies

Table 2: Common Causes of HIGH INRs and Management Strategies

***Most common drugs that can increase INR:**

- Antibiotics: sulfamethoxazole/trimethoprim, metronidazole, quinolones (ciprofloxacin, levofl erythromycin, clarithromycin, azithromycin
- Azole antifungals: fluconazole, voriconazole
- Cardiac drugs: amiodarone, some statins (atorvastatin and pravastatin are least likely to interact), fenofibrate
- Acetaminophen >1 g/day
- Levothyroxine dose changes– full effect observed after 4-6 weeks of dose change

Other Relevant

- Warfarin
- Warfarin: Point-of-Care INR Monitoring of Warfarin Therapy

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References:

Ageno W, et al. Oral anticoagulant therapy: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 Suppl):e44S-e88S.

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Smythe MA, et al. Anticoagulant drug-drug interactions with cannabinoids: A systematic review. Pharmacotherapy. 2023;43(12):1327-1338.

Witt DM, et al. American Society of Hematology 2018 guidelines for management of venous thromboembolism: optimal management of anticoagulation therapy. Blood Adv. 2018;2(22):3257–3291.

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